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The lattice Boltzmann method for fluid dynamics

14.1 Introduction

Flow problems are widely studied in engineering because of their relevance to industrial processes and environmental problems. Such problems belong to the realm of macroscopic phenomena which are formulated in terms of one or more, possibly nonlinear, partial differential equations. If there is no possibility of exploiting symmetry, allowing for separation of variables, these equations are usually solved using finite element or finite difference methods.

The standard problem is the flow of a one-component, isotropic nonpolar liquid, which is described by the Navier–Stokes equations. These equations are based on mass and momentum conservation, and on the assumption of isotropic relaxation towards equilibrium. Finite element methods have been described in [Chapter 13](#) for elasticity; the methods described there may be extended and adapted to develop codes for computational fluid dynamics (CFD), which are widely used by engineers. Such an extension is beyond the scope of this book.

A finite element solution of the Navier–Stokes equations may sometimes become cumbersome when the boundary conditions become exceptionally complicated, as is the case with flow through porous media where the pore diameter becomes very small (and the number of pores very large). Other cases where finite element methods run into problems are multiphase or binary systems, where two different substances or phases exist in separate regions of space. These regions change their shape and size in the course of time. Usually, the finite element points are taken on the system boundaries, but that implies complicated bookkeeping, in particular when the topology of the regions changes, as is the case in coalescence of droplets.

A possible solution to these problems characterized by difficult topologies is to use regular, structured grids, and let the interfaces move over the grid. In that case, we need special couplings for nearest neighbour grid points that are separated by the interface (such as a phase boundary). This ‘immersed interface’ method has been

pioneered by LeVeque and Li [1]. Another method is to replace the macroscopic, finite element approach by a microscopic particle approach, such as molecular dynamics, but this usually requires so many particles that it is no longer feasible to do useful simulations. Since the Navier–Stokes equations are rather universal in that they only include two viscosities and the mass density as essential parameters, it seems that the details of the interactions between the particles do not all matter: only few features of these interactions will survive in the macroscopic description. This is in some sense similar to the macroscopic description of elasticity, where the elasticity tensor, which is based on two elastic or Lamé constants, is the only fingerprint surviving from the microscopic details of the interactions.

This suggests that an alternative approach may be to use a ‘mock fluid’, consisting of ‘mock particles’ with very simple microscopic properties which are tuned to yield the correct hydrodynamic behaviour. This scheme is adopted in the *lattice Boltzmann* approach. The particles are put on a lattice and can move only to neighbouring lattice sites. Interactions between the particles and relaxation effects are included, and the resulting system yields hydrodynamic behaviour on large scales. This method has been applied successfully to binary flows, complex geometries, objects moving in a flow, etc.

We start this chapter by deriving the Navier–Stokes equations from the Boltzmann equation in the continuum (Section 14.2). Then we formulate the Boltzmann approach on a discrete lattice (Section 14.3) and consider an example. In Section 14.4, we apply the method to binary systems. Finally, in Section 14.5, we show that the lattice Boltzmann model leads in some limit to the Navier–Stokes equations for fluids in the incompressible limit. For more information about the method and its applications, the interested reader may consult Refs. [2–4].

14.2 Derivation of the Navier–Stokes equations

In this section we present a derivation of the Navier–Stokes equations from an approximate Boltzmann equation through a Chapman–Enskog procedure [5]. This works as follows. We start by defining the particle distribution function $n(\mathbf{r}, \mathbf{v}, t)$ which gives us the number density of particles with velocity $\mathbf{v}$ inside a small cell located at $\mathbf{r}$, at time t . This distribution for $\mathbf{r}$ and $\mathbf{v}$ will change in time because particles have a velocity and therefore move to a different position, and because the particles collide, which results in exchange of momentum and energy. The evolution of the distribution function is described by the well-known *Boltzmann equation* which describes a dilute system. The picture behind the Boltzmann equation is that of particles moving undisturbed through phase space most of the time, but experiencing every now and then a collision with other particles, and these collisions are considered to be instantaneous. The Boltzmann equation works very well, even

in cases where the substance is not so dilute, such as a fluid. The Boltzmann equation reads:

$$\frac{\partial n}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} n = \left(\frac{dn}{dt} \right)_{\text{collisions}}. \quad (14.1)$$

The second term describes the change due to particle flow, and the right hand side is the change due to collisions.

If the particles were simply to flow according to their initial velocity, without interaction, equilibrium would never be reached: the role of the collisions is to establish *local equilibrium*, that is, a distribution which is in equilibrium in a small cell with fixed volume, constant temperature, density and average velocity $\mathbf{u}$. We know this equilibrium distribution; it is given as

$$n^{\text{eq}}(\mathbf{r}, \mathbf{v}) = \left(\frac{m\pi}{2k_{\text{B}}T} \right)^{3/2} n(\mathbf{r}) \exp\{-m[\mathbf{v} - \mathbf{u}(\mathbf{r})]^2/(2k_{\text{B}}T)\}, \quad (14.2)$$

which holds for cells small enough to justify a constant potential. Once the liquid is in (local) equilibrium, the collisions will not push it away from equilibrium. It can be shown that the collisions have the effect of increasing the entropy – hence they generate heat.

Before we continue, we note that the mass must *always* be conserved, whether there are collisions or not. The mass density is found as

$$\rho(\mathbf{r}, t) = \int m n(\mathbf{r}, \mathbf{v}, t) d^3v = m n(\mathbf{r}). \quad (14.3)$$

Its evolution can be calculated by integrating equation (14.1), multiplied by the single particle mass m , over the velocity:

$$\frac{\partial \rho(\mathbf{r}, t)}{\partial t} + \int m \mathbf{v} \cdot \nabla_{\mathbf{r}} n(\mathbf{r}, \mathbf{v}, t) d^3v = \int \left(m \frac{dn}{dt} \right)_{\text{collisions}} d^3v. \quad (14.4)$$

The second term of this equation can be written as $\nabla \cdot \mathbf{j}(\mathbf{r}, t)$ where $\mathbf{j}$ denotes the mass flux, or momentum density, of the fluid:

$$\mathbf{j}(\mathbf{r}, t) = \int m \mathbf{v} n(\mathbf{r}, \mathbf{v}, t) d^3v = \rho \mathbf{u}, \quad (14.5)$$

where $\mathbf{u}$ is the average local velocity. This equation can be checked using (14.2). The collisions change the velocity distribution, but not the mass density of the particles. Hence the right hand side of (14.4) vanishes and we obtain the familiar continuity equation:

$$\frac{\partial \rho(\mathbf{r}, t)}{\partial t} + \nabla \cdot \mathbf{j}(\mathbf{r}, t) = 0. \quad (14.6)$$

Another interesting equation describes the conservation of momentum. We would like to know how $\mathbf{j}(\mathbf{r}, t)$ changes with time. This is again evaluated straightforwardly

by multiplying Eq. (14.1) by $\mathbf{v}$ and integrating over the velocity. Using the indices α and β for the Cartesian coordinates, we obtain

$$\frac{\partial j_\alpha}{\partial t} + \int mv_\alpha \sum_\beta v_\beta \partial_\beta n(\mathbf{r}, \mathbf{v}, t) d^3v = \int mv_\alpha \left(\frac{dn}{dt} \right)_{\text{collisions}} d^3v, \quad (14.7)$$

where ∂_β denotes a derivative with respect to the coordinate r_β . For the right hand side, a similar statement can be made as for the equivalent term in the mass equation: although individual particles involved in a collision change their momenta, the *total* momentum is conserved on collision. After thus putting the right hand side to zero, we write (14.7) in shorthand notation as

$$\frac{\partial j_\alpha}{\partial t} + \partial_\beta \Pi_{\alpha\beta} = 0, \quad (14.8)$$

where we have introduced the momentum flow tensor

$$\Pi_{\alpha\beta} = \int mv_\alpha v_\beta n(\mathbf{r}, \mathbf{v}, t) d^3v, \quad (14.9)$$

and where we have used the Einstein summation convention in which repeated indices (in this case β) are summed over.

Assuming that we are in equilibrium, we can evaluate the momentum tensor by substituting for $n(\mathbf{r}, \mathbf{v}, t)$ the form (14.2):

$$\begin{aligned} \Pi_{\alpha\beta}^{\text{eq}} &= \int v_\alpha v_\beta \rho(\mathbf{r}) \exp[-m(\mathbf{v} - \mathbf{u})^2 / (2k_B T)] d^3v \\ &= \rho(\mathbf{r}) (k_B T \delta_{\alpha\beta} + u_\alpha u_\beta). \end{aligned} \quad (14.10)$$

This result can be derived by separately considering $\alpha = \beta$ and $\alpha \neq \beta$, and working out the appropriate Gaussian integrals (using the substitution $\mathbf{w} = \mathbf{v} - \mathbf{u}$). Noting that $\rho k_B T$ equals the pressure P ,¹ we arrive at the following two equations:

$$\frac{\partial \rho(\mathbf{r}, t)}{\partial t} + \nabla \cdot \mathbf{j}(\mathbf{r}, t) = 0 \text{ (mass conservation);} \quad (14.11a)$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla_{\mathbf{r}} \cdot (P \mathbf{I} + \rho \mathbf{u} \mathbf{u}) = 0 \text{ (momentum conservation).} \quad (14.11b)$$

Using the first equation, we can rewrite the second as

$$\frac{\partial \mathbf{u}(\mathbf{r}, t)}{\partial t} + [\mathbf{u}(\mathbf{r}, t) \cdot \nabla_{\mathbf{r}}] \mathbf{u}(\mathbf{r}, t) = -\frac{1}{\rho(\mathbf{r}, t)} \nabla_{\mathbf{r}} P(\mathbf{r}, t). \quad (14.12)$$

The equations (14.11a) and (14.11b) or (14.12) are the *Euler equations* for a fluid in equilibrium. These equations neglect dissipative effects.

When the fluid is not everywhere in local equilibrium, the collisions will drive the system towards equilibrium – hence their effect can no longer be neglected.

¹ Here, we consider the fluid as an ideal gas; a realistic equation of state may be used instead.

As mentioned above, the additional currents which arise on top of the equilibrium currents increase the entropy and are therefore called *dissipative*. Hence these terms describe the viscous effects in the fluid.

We now split the distribution function into an equilibrium and a nonequilibrium part:

$$n(\mathbf{r}, \mathbf{v}, t) = n^{\text{eq}}(\mathbf{r}, \mathbf{v}) + n^{\text{noneq}}(\mathbf{r}, \mathbf{v}, t). \quad (14.13)$$

The equilibrium term satisfies (14.2).

How can we represent the effect of the collision term? There is an approach due to Maxwell, which is based on the assumption that all relaxation processes have the same, or are dominated by a single, relaxation time τ . In that case:

$$\left(\frac{dn(\mathbf{r}, \mathbf{v}, t)}{dt} \right)_{\text{collisions}} = - \frac{n(\mathbf{r}, \mathbf{v}, t) - n^{\text{eq}}(\mathbf{r}, \mathbf{v})}{\tau} = - \frac{n^{\text{noneq}}}{\tau}. \quad (14.14)$$

As mentioned above, the collisions do not change the mass conservation equation, which should always be valid. The equation for the flux will, however, acquire a contribution from the nonequilibrium part of the distribution function, as we shall see. The mass flux can still be written as $\rho \mathbf{u}$. Moreover, the collisions leave the total momentum unchanged.

The flux $\mathbf{j}$ occurring in the mass conservation equation also occurs in the momentum conservation equation. In this second equation, the momentum flux $\Pi_{\alpha\beta}$ occurs, which we have calculated above *assuming equilibrium*. If we consider the evolution of this flux using the Boltzmann equation, we see that the collision effects enter explicitly in this momentum flux.

To find the lowest-order contribution to a systematic expansion of the density, we replace n on the left hand side of the Boltzmann equation by its equilibrium version:

$$\frac{\partial n^{\text{eq}}(\mathbf{r}, \mathbf{v})}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{r}} n^{\text{eq}} = - \frac{n^{\text{noneq}}(\mathbf{r}, \mathbf{v}, t)}{\tau}. \quad (14.15)$$

This is an *explicit* equation for the nonequilibrium term. It can be shown that this is an expansion in the parameter ℓ/L , where ℓ is the mean free path, and L is the typical length scale over which the hydrodynamic quantities vary [6]. Note that if we integrate this equation over the velocity, the right hand side vanishes as the collisions do not affect the mass density.

The momentum flux is defined in (14.9). This is calculated from the density $n(\mathbf{r}, \mathbf{v}, t)$ and it can therefore be split into an equilibrium and a nonequilibrium part. The equilibrium part was calculated in Eq. (14.10), and the nonequilibrium part

will now be calculated using (14.15):

$$\begin{aligned}\Pi_{\alpha\beta}^{\text{noneq}} &= \int m v_\alpha v_\beta n^{\text{noneq}} d^3v \\ &= -\tau \left[\int m v_\alpha v_\beta \frac{\partial n^{\text{eq}}}{\partial t} d^3v + \int m v_\alpha v_\beta \mathbf{v} \cdot \nabla_{\mathbf{r}} n^{\text{eq}} d^3v \right].\end{aligned}\quad (14.16)$$

Before we proceed to work out (14.16) further, we note that the tensor $\Pi_{\alpha\beta}^{\text{noneq}}$ has an important property: its trace vanishes. This can be seen by writing out this trace:

$$\sum_{\alpha} \Pi_{\alpha\alpha}^{\text{noneq}} = \int v^2 n^{\text{noneq}}(\mathbf{r}, \mathbf{v}, t) d^3v. \quad (14.17)$$

Realizing that this expression represents the change in the average kinetic energy due to the collisions, we immediately see that it vanishes as the (instantaneous) collisions leave the total energy invariant:

$$\text{Tr } \Pi^{\text{noneq}} = 0. \quad (14.18)$$

For the calculation of the nonequilibrium stress tensor, Eq. (14.16), we use the following equations, which can easily be seen to hold for the equilibrium distribution:

$$\int m n^{\text{eq}}(\mathbf{r}, \mathbf{v}) d^3v = \rho(\mathbf{r}); \quad (14.19a)$$

$$\int m(v_\alpha - u_\alpha)(v_\beta - u_\beta) n^{\text{eq}}(\mathbf{r}, \mathbf{v}) d^3v = \rho \frac{k_B T}{m} \delta_{\alpha\beta} = P \delta_{\alpha\beta}; \quad (14.19b)$$

$$\dot{u}_\alpha = - \sum_{\beta} u_\beta \partial_\beta u_\alpha - \frac{1}{\rho} (\partial_\alpha P); \quad (14.19c)$$

where in the last equation it is understood that the velocities are those evaluated for the equilibrium distribution: this equation is the Euler equation, (14.12).

We first work out the first term in the square brackets on the right hand side in (14.16), which can also be written as $\partial_t \Pi_{\alpha\beta}^{\text{eq}}$ (we use ∂_t to denote a derivative with respect to t). After some manipulation, using Eqs. (14.10), (14.11a) and (14.11b) (or (14.19c)), this can be written as

$$\begin{aligned}\partial_t \Pi_{\alpha\beta}^{\text{eq}} &= \partial_t (P \delta_{\alpha\beta} + \rho u_\alpha u_\beta) \\ &= \dot{P} \delta_{\alpha\beta} - \sum_{\gamma} [\partial_\gamma (\rho u_\gamma) u_\alpha u_\beta + \rho u_\alpha u_\gamma (\partial_\gamma u_\beta) + \rho u_\beta u_\gamma (\partial_\gamma u_\alpha)] \\ &\quad - u_\beta \partial_\alpha P - u_\alpha \partial_\beta P.\end{aligned}\quad (14.20)$$

The second term in the square brackets of (14.16) can be written, using the quantity $w_\alpha = v_\alpha - u_\alpha$ (see also Eq. (14.10) and (14.19b)), in the form

$$\begin{aligned} & \int (u_\alpha + w_\alpha)(u_\beta + w_\beta)(u_\gamma + w_\gamma) \partial_\gamma n^{\text{eq}}(\mathbf{r}, \mathbf{v}) \, d^3v \\ &= \partial_\gamma (\rho u_\alpha u_\beta u_\gamma + u_\alpha P \delta_{\beta\gamma} + u_\beta P \delta_{\alpha\gamma} + u_\gamma P \delta_{\alpha\beta}). \end{aligned} \quad (14.21)$$

This term can now be worked out to yield

$$\begin{aligned} & \sum_\gamma [u_\alpha u_\beta u_\gamma \partial_\gamma \rho + \rho u_\beta u_\gamma (\partial_\gamma u_\alpha) + \rho u_\alpha u_\gamma (\partial_\gamma u_\beta) + \rho u_\alpha u_\beta (\partial_\gamma u_\gamma) \\ &+ \partial_\gamma (P u_\gamma) \delta_{\alpha\beta} + \partial_\gamma (P u_\alpha) \delta_{\beta\gamma} + \partial_\gamma (P u_\beta) \delta_{\alpha\gamma}]. \end{aligned} \quad (14.22)$$

Adding the two terms of Eq. (14.16), many terms occurring in (14.20) and (14.22) cancel. The ones that remain are

$$P(\partial_\beta u_\alpha + \partial_\alpha u_\beta) + \delta_{\alpha\beta} \left\{ \dot{P} + \sum_\gamma [u_\gamma (\partial_\gamma P) + P \partial_\gamma u_\gamma] \right\}. \quad (14.23)$$

The terms

$$\dot{P} + \sum_\gamma u_\gamma (\partial_\gamma P) \quad (14.24)$$

can be calculated using (14.19b) and the equilibrium distribution. When we write this out, we obtain, again with $w_\alpha = v_\alpha - u_\alpha$:

$$\begin{aligned} & \frac{\partial}{\partial t} \int m w^2 n \, d^3w + \sum_\gamma u_\gamma \partial_\gamma \int m w^2 n \, d^3w \\ &= \int m w^2 \left(\frac{\partial n}{\partial t} + \sum_\gamma u_\gamma \partial_\gamma n \right) d^3w = -\frac{1}{\tau} \int m w^2 n^{\text{noneq}} \, d^3w. \end{aligned} \quad (14.25)$$

This is the trace of the tensor

$$-\frac{1}{\tau} \int m w_\alpha w_\beta n^{\text{noneq}} \, d^3v. \quad (14.26)$$

Now we use the fact that $\text{Tr} \Pi^{\text{noneq}}$ vanishes. This can only happen when the trace occurring in the last equation cancels the trace of the remaining terms in the expression for Π^{noneq} . This tensor must therefore be

$$\Pi^{\text{noneq}} = -P\tau \left(\partial_\alpha u_\beta + \partial_\beta u_\alpha - \frac{2}{3} \delta_{\alpha\beta} \partial_\gamma u_\gamma \right). \quad (14.27)$$

Using this, we can formulate the momentum conservation equation, with $\nu = \tau k_B T / m$, as

$$\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} = -\frac{1}{\rho} \nabla P + \nu \nabla^2 \mathbf{u} + \frac{1}{3} \nu \nabla (\nabla \cdot \mathbf{u}). \quad (14.28)$$

The mass conservation equation and the momentum conservation equation together are insufficient to give us the four unknown field: ρ , u_x , u_y and P . We therefore need an additional equation, which may be $\rho = \text{constant}$ for an incompressible fluid, or $P \propto \rho$ for the isothermal case (which we have analysed here). Note that the case where $\rho = \text{constant}$ also implies $\nabla \cdot \mathbf{u} = 0$ from the continuity equation, which in turn causes the last term in the last equation to become negligible.

14.3 The lattice Boltzmann model

The idea of the lattice Boltzmann method is to use a ‘toy model’ for the liquid, which combines the properties of mass, momentum and energy conservation (only the first two are used in the derivation of the fluid equations) and isotropic relaxation of the stress. In the simplest case, the Maxwell *Ansatz* of a single dominant relaxation time is used. Some time ago, the *lattice gas* model was studied intensively [7–9]. In this model, the fluid consists of particles which can occupy lattice sites. In two dimensions, the lattice can be square (with links to nearest and to next nearest, diagonal neighbours) or hexagonal. The time is also taken discrete, and at subsequent time steps, particles are allowed to move to a neighbouring position, according to rules which guarantee mass and momentum conservation. Because of the discrete nature of the model, the fluctuations in this method are substantial, and an alternative formulation was developed: the lattice Boltzmann model. This is also formulated on a lattice, and the time is discrete, but the particles are replaced by *densities* on the lattice sites.

Let us specify the formulation on a hexagonal lattice. This is shown on the left hand side of [Figure 14.1](#). The arrows indicate possible velocities of particles at each site: the particles with one of these velocities travel in one time step to the neighbour at the other end of the link. On the hexagonal lattice, the velocities are all equal in size: one lattice constant per time step. Particles can also stand still. In a lattice gas cellular automaton, individual particles are distinguished; in the lattice Boltzmann method, however, we only consider *densities* of particles with a particular velocity: n_i is the density of particles with velocity $\mathbf{v}_i$, which may be directed along a lattice link or may be 0.

To be more specific, we have seven possible velocities at each link (including the zero-velocity particles), and therefore we have seven densities, one for each velocity. At each time step, two processes take place, which can be directly related to the Boltzmann equation:

- At each site, all particles with a nonzero speed move to the neighbouring site to which their velocity points (see the description above);
- At each site, the distribution is relaxed to the equilibrium distribution.

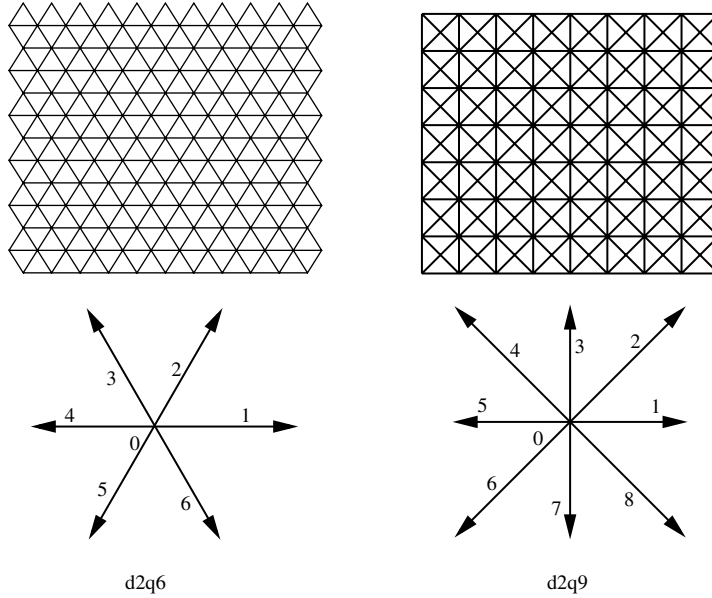


Figure 14.1. The two grids used in two-dimensional lattice Boltzmann simulations. On the left hand side is the hexagonal grid (d2q6; see text) with the possible particle velocities. On the right hand side, the d2q9 grid is shown. A square grid is not suitable, as on such a grid all momenta along each horizontal and each vertical grid line would be conserved.

The second step represents the effect of the collisions.

This algorithm is in fact a realisation of the following equation which governs the evolution of the distribution in time:

$$n_i(\mathbf{r} + c\Delta t \mathbf{e}_i, t + \Delta t) = n_i(\mathbf{r}, t) - \frac{1}{\tau} [n_i(\mathbf{r}, t) - n_i^{\text{eq}}(\mathbf{r}, t)]. \quad (14.29)$$

Here, i labels the seven possible directions $\mathbf{e}_i$ from each site ($i = 0$ represents the rest particles), and n^{eq} is the equilibrium distribution. Furthermore, c is related to the velocity at which the particles with $i \neq 0$ move; in practice we take c and Δt both equal to 1 in the end, but for derivations it is useful to keep them as parameters in the problem.

The first step can easily be carried out: for each site, we loop over the six nonzero speeds, and move the particles to the neighbouring site. Care must be taken to distinguish those which have not yet been moved from the ‘newcomers’, to avoid the latter being moved twice. The relaxation is then straightforward, provided we know the equilibrium distribution. We can construct this equilibrium distribution as a second order polynomial in terms of the velocity components u_α :

$$n_i^{\text{eq}}(\mathbf{u}) = n_i^{\text{eq}}(0)(1 + Au_\alpha e_{i\alpha} + Bu_\alpha u_\alpha + Cu_\alpha u_\beta e_{i\alpha} e_{i\beta}), \quad (14.30)$$

where α and β are Cartesian coordinates x and y ; the dependence on the positions is left out for convenience. All terms in this expression have been chosen to be isotropic, i.e. there is no preferred direction.

An additional condition is Gallilean invariance: all moments

$$M^j(\mathbf{u}) = \sum_{i=0}^6 (\mathbf{u} - \mathbf{e}_i)^j n_i^{\text{eq}} \quad (14.31)$$

should satisfy $M^j(\mathbf{u}) = M(0)$, for $j = 0, 1, 2, \dots$. After some calculation, this condition leaves [10, 11]

$$n_i^{\text{eq}}(\mathbf{u}) = w_i \frac{\rho}{m} \left(1 + \frac{4}{c^2} u_\alpha e_{i\alpha} - \frac{2}{c^2} u_\alpha u_\alpha + \frac{8}{c^4} u_\alpha u_\beta e_{i\alpha} e_{i\beta} \right) \quad (14.32)$$

as the only possibility for the equilibrium distribution. In (14.32), an implicit summation over repeating Cartesian indices (α, β) is understood. Furthermore,

$$w_0 = 1/2; \quad (14.33a)$$

$$w_i = 1/12, \quad i = 1, \dots, 6. \quad (14.33b)$$

For a square lattice, we should allow the particles to visit the four neighbours along the two Cartesian directions, and the four next-nearest neighbours as shown in Figure 14.1. This is called the ‘d2q9’ model because it is two-dimensional and at each site there are nine possible velocities. For this model, the equilibrium distribution can be derived similarly – the result is:

$$n_i^{\text{eq}} = w_i \frac{\rho}{m} \left(1 + \frac{3}{c^2} e_{i\alpha} u_\alpha + \frac{9}{2c^4} e_{i\alpha} e_{i\beta} u_\alpha u_\beta - \frac{3}{2} \frac{u_\alpha u_\alpha}{c^2} \right). \quad (14.34)$$

Here,

$$w_i = \begin{cases} 4/9 & \text{for } i = 0; \\ 1/9 & \text{for } i = 2, 4, 6, 8; \\ 1/36 & \text{for } i = 1, 3, 5, 7. \end{cases} \quad (14.35)$$

We now have all the ingredients to build the lattice Boltzmann program, if we know how to handle the boundary conditions. In fluid dynamics, the boundary condition at a wall is usually defined to correspond to zero velocity (this is called a ‘stick’ boundary condition as opposed to ‘slip’ boundary conditions). The simplest scheme to realise this is to use the ‘bounce back’ boundary condition. This assumes that boundaries lie in between neighbouring grid points. For a point lying just inside the system, we move its particles pointing to a neighbouring point outside the system to that neighbour and reverse its velocity there. On average, this boils down

to having zero velocity in between the two particles. More accurate implementations for the boundary conditions have been developed [12, 13].

In Section 14.5 we show that the lattice Boltzmann model leads to the Navier–Stokes equations for an incompressible fluid in the limit of small velocities. There, it will be shown that the only parameter of the algorithm, which is the relaxation time τ , is related to the viscosity ν by

$$\nu = \frac{2\tau - 1}{6} \frac{\Delta x^2}{\Delta t}, \quad (14.36)$$

where $\Delta x, \Delta t$ are the lattice constant and the time steps (which are usually taken equal to 1).

PROGRAMMING EXERCISE

Construct a lattice Boltzmann code for the flow through a two-dimensional pipe which we imagine to be horizontal. This is a rectangle, where on the left hand side we supply fluid, which is drained on the right hand side. Use the d2q9 lattice. On the lateral boundaries, the bounce back rule is used to ensure stick boundary conditions. The easiest way to realise the flow is by imposing a pressure gradient over the system from left to right. This means that on each segment of the fluid (corresponding to a point of the Boltzmann lattice), a constant force is acting. This has the effect of increasing all velocities along the direction of the flow at each time step by the same (small) amount.

The algorithm is set up as follows.

- Move the density n_i to the appropriate neighbour;
- Reverse the new velocities on points beyond the system boundaries;
- Calculate velocities at each point;
- Add a small velocity along the direction of the pressure gradient;
- Calculate equilibrium distribution at each point;
- Relax the densities at the points inside the system according to

$$n_i^{\text{new}} = (1 - 1/\tau)n_i^{\text{old}} + n_i^{\text{eq}}/\tau.$$

Check If your program works correctly, you should obtain a parabolic flow profile throughout the pipe. Note that the simulation is only reliable for small velocities. The parabola should have curvature $\nabla P/(\rho\nu)$. The viscosity ν is related to the relaxation time τ as $\nu = (2\tau - 1)/6$ (see the next section). See also Problem 14.1.

14.4 Additional remarks

The lattice Boltzmann method works on a hexagonal lattice and on a square lattice provided the stresses can relax isotropically. This requirement forces us

to include the next nearest neighbours into the possible moves on the square lattice. In three dimensions, a so-called FCHC lattice must be used [7, 8], which contains moves to the neighbours along Cartesian directions, as well as moves to neighbours at relative positions $(1, 1, 0)$, $(1, 0, -1)$ etc. We shall not go further into this; an extensive literature exists on the three-dimensional version of the model (see most references in this chapter).

An interesting aspect of the lattice Boltzmann method is that it can easily be extended to problems that are usually difficult to treat using other methods. These include binary or multiphase systems and objects moving in the flow. Here we concentrate on binary systems. Two methods are predominant in this field: the first was proposed by Shan and Chen [14] and the second by Swift *et al.* [15]. We shall adopt the first method here for its simplicity. Shan and Chen start by simulating two fluids in parallel. These fluids can be conveniently be denoted by a ‘colour’: say, red and blue particles. In principle, the fluids may have different viscosities (thus different values for the relaxation times τ) – we shall take them to be equal in our description. The two fluids interact through some potential which has the form

$$V(\mathbf{r}, \mathbf{r}') = G_{cc'}(\mathbf{r} - \mathbf{r}') \rho_c(\mathbf{r}) \rho_{c'}(\mathbf{r}'). \quad (14.37)$$

The indices c and c' denote the colours and $\rho_c = \sum_i m_c n_{c,i}$. The kernel $G_{cc'}$ is zero for equal colours $c = c'$ (which assume the values r and b for red and blue respectively). Furthermore, $G_{rb}(\mathbf{r}, \mathbf{r}')$ is only nonzero when $\mathbf{r}$ and $\mathbf{r}'$ are lattice neighbours.

The average velocities of the fluids are calculated for the two fluids together:

$$\mathbf{u} = \frac{m_r \sum_i \mathbf{e}_i n_{r,i} + m_b \sum_i \mathbf{e}_i n_{b,i}}{\sum_i (m_r n_{r,i} + m_b n_{b,i})}. \quad (14.38)$$

The relaxation of the distributions at each site is with respect to the equilibrium distribution based on this average velocity.

From this potential we can derive a force by taking the (discrete) gradient. This directly leads to an extra force on a particle with colour c located at $\mathbf{r}$ of the form

$$\frac{dp_c(\mathbf{r})}{dt} = -\rho_c(\mathbf{r}) \sum_i G_{cc',i} \rho_{c'}(\mathbf{r} + \mathbf{e}_i) \quad (14.39)$$

where $c \neq c'$. The interaction $G_{cc',i}$ assumes different values for nearest (nn) and next nearest neighbours (nnn) respectively. They must be tuned to make the force isotropic – in our case this means

$$G_{nnn} = \sqrt{2} G_{nn}. \quad (14.40)$$

The extra force is included in the lattice Boltzmann equation which now reads

$$n_{c,i}(\mathbf{r} + \mathbf{e}_i, t + 1) = n_{c,i}(\mathbf{r}, t) - \frac{1}{\tau} [n_{c,i}(\mathbf{r}, t) - n_i^{\text{eq}}(\mathbf{r}, t)] + \frac{dp_c}{dt}(\mathbf{r}, t). \quad (14.41)$$

In summary, the heart of the algorithm reads (the change in velocity related to the force dp_c/dt is called Δv):

```

Calculate average velocities from (14.38);
FOR each site  $\mathbf{r}_1$  DO
  Calculate local densities  $\rho_1(c)$ 
    for each colour  $c$ ;
  FOR each neighbour  $\mathbf{r}_2$  of  $\mathbf{r}_1$  DO
    Calculate density  $\rho_2(c')$  of that neighbour;
    Subtract  $\Delta v \rho_2(c')$  from the velocity of colour  $c$  at  $\mathbf{r}_1$ ;
  END FOR;
END FOR.
```

You must first put the two colours on each site with a fixed concentration ratio of, for example, 2 to 1. It turns out that for high values of Δv , the simulation becomes unstable. This is due to the fact that when the different colours repel each other too strongly, we get excessively high velocities, and this results in negative densities. For $\tau = 1$ the divergence sets in for $\Delta v > 0.11$ on the d2q9 grid (for a concentration ratio of 2:1). The instability is due to the speed exceeding the sound speed.

Now the simulation is fully defined and can be implemented (see Problem 14.3, which also addresses the analysis of the data). The result shows the formation of bubbles which grow by coalescence. The bubbles should satisfy Laplace's law, which states that the jump in pressure when going from inside the bubble to the outside should be proportional to the inverse of the radius of the bubble [16].

Figure 14.2 shows the pressure drop for different bubbles as a function of inverse radius. Clearly, our simulation satisfies Laplace's law.

*14.5 Derivation of the Navier–Stokes equation from the lattice Boltzmann model

In this section, we shall show how the Navier–Stokes equations can be recovered in the incompressible limit from the lattice Boltzmann model. The derivation is based on two major ingredients [15, 17, 18]:

- A systematic expansion of $n_i(\mathbf{r}, t)$ in the time step Δt is made;
- Terms of the form $(u/c_s)^j$ are neglected beyond a certain order j . The quantity u/c_s , where c_s is the sound speed, is known as the *Mach number*, M .

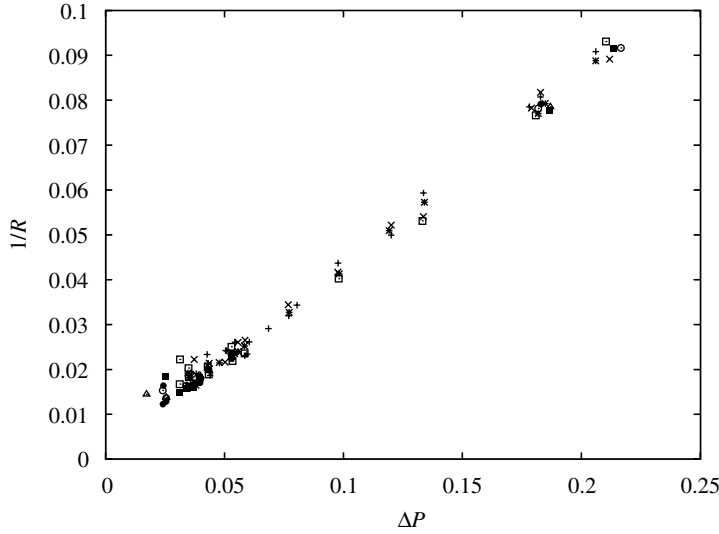


Figure 14.2. Pressure drop across droplet boundary. Data are taken from a long run: different symbols correspond to different times. The points are seen to fall on a straight line. In the simulation we have taken $\tau = 1$, and $\Delta\nu = 0.085$. The times vary between 6000 and 14 000 steps.

We start with the systematic expansion of n in powers of Δt . Taylor-expanding Eq. (14.29), we have

$$-\frac{n_i(\mathbf{r}, t) - n_i^{\text{eq}}(\mathbf{r}, t)}{\tau} = \sum_{k=0}^{\infty} \frac{(\Delta t)^k}{k!} D_i^k n_i(\mathbf{r}, t) \quad (14.42)$$

where we have introduced the total differential operator

$$D_i = \frac{\partial}{\partial t} + c e_{i\alpha} \partial_\alpha. \quad (14.43)$$

We call $n_i^{(k)}(\mathbf{r}, t)$ an approximation of $n_i(\mathbf{r}, t)$ which is correct to order k in Δt . We then have:

$$n_i^{(0)}(\mathbf{r}, t) = n_i^{\text{eq}}(\mathbf{r}, t) \quad (14.44a)$$

$$n_i^{(1)}(\mathbf{r}, t) = n_i^{\text{eq}}(\mathbf{r}, t) - \tau \Delta t D_i n_i^{\text{eq}}(\mathbf{r}, t) \quad (14.44b)$$

$$n_i^{(2)}(\mathbf{r}, t) = n_i^{\text{eq}}(\mathbf{r}, t) - \tau \Delta t D_i n_i^{(1)}(\mathbf{r}, t) - \tau \frac{\Delta t^2}{2} D_i^2 n_i^{\text{eq}}(\mathbf{r}, t). \quad (14.44c)$$

Substituting the second equation for $n_i^{(1)}$ into the third, we obtain

$$\begin{aligned} n_i^{(2)}(\mathbf{r}, t) &= n_i^{\text{eq}}(\mathbf{r}, t) - \tau \Delta t (\partial_t + c e_{i\alpha} \partial_\alpha) n_i^{\text{eq}}(\mathbf{r}, t) \\ &\quad - \tau \Delta t^2 \left(\frac{1}{2} - \tau \right) (\partial_t + c e_{i\alpha} \partial_\alpha)^2 n_i^{\text{eq}}(\mathbf{r}, t). \end{aligned} \quad (14.45)$$

Now we consider the moments of these equations. The zeroth moment of the nonequilibrium and equilibrium distribution gives us the density:

$$\sum_i mn_i = \sum_i mn_i^{\text{eq}} = \rho \quad (14.46)$$

and the first moment the mass flux:

$$\sum_i mce_{i\alpha}n_i = \sum_i mce_{i\alpha}n_i^{\text{eq}} = j_\alpha \quad (14.47)$$

We furthermore define the second and third moments:

$$\sum_i mc^2 e_{i\alpha} e_{i\beta} n_i \equiv \Pi_{\alpha\beta} \quad (14.48)$$

and

$$\sum_i mc^3 e_{i\alpha} e_{i\beta} e_{i\gamma} n_i \equiv \Gamma_{\alpha\beta\gamma} \quad (14.49)$$

We now take the zeroth moment of [Eq. \(14.45\)](#). Realising that this moment is identical for n_i^{eq} and $n_i^{(2)}$, we obtain

$$\partial_t \rho + \partial_\alpha j_\alpha = \left(\tau - \frac{1}{2}\right) \Delta t (\partial_t^2 \rho + 2\partial_t \partial_\alpha j_\alpha + \partial_\alpha \partial_\beta \Pi_{\alpha\beta}^{(0)}) + \mathcal{O}(\Delta t^2). \quad (14.50)$$

For the first moment, we obtain

$$\partial_t j_\alpha + \partial_\beta \Pi_{\alpha\beta}^{(0)} = \left(\tau - \frac{1}{2}\right) \Delta t (\partial_t^2 j_\alpha + 2\partial_t \partial_\beta \Pi_{\alpha\beta}^{(0)} + \partial_\beta \partial_\gamma \Gamma_{\alpha\beta\gamma}^{(0)}) + \mathcal{O}(\Delta t^2). \quad (14.51)$$

Now we note that the $\mathcal{O}(\Delta t)$ term on the right hand side of the [Eq. \(14.50\)](#) can be written as

$$\partial_t^2 \rho + 2\partial_t \partial_\alpha j_\alpha + \partial_\alpha \partial_\beta \Pi_{\alpha\beta}^{(0)} = \partial_t (\partial_t \rho + \partial_\alpha j_\alpha) + \partial_\alpha (\partial_t j_\alpha + \partial_\beta \Pi_{\alpha\beta}^{(0)}). \quad (14.52)$$

From [Eqs. \(14.50\)](#) and [\(14.51\)](#), the two terms on the right hand side of this equation can easily be seen to be of order Δt , so we see that

$$\partial_t \rho + \partial_\alpha j_\alpha = 0 + \mathcal{O}(\Delta t^2). \quad (14.53)$$

We have now recovered the continuity equation to order Δt^2 .

For the other moment equation, [\(14.51\)](#), we can argue along the same lines that the first half of the $\mathcal{O}(\Delta t)$ term is close to zero, and we are left with

$$\partial_t j_\alpha + \partial_\beta \Pi_{\alpha\beta}^{(0)} = \left(\tau - \frac{1}{2}\right) \Delta t (\partial_t \partial_\beta \Pi_{\alpha\beta}^{(0)} + \partial_\beta \partial_\gamma \Gamma_{\alpha\beta\gamma}^{(0)}) + \mathcal{O}(\Delta t^2). \quad (14.54)$$

To proceed, we must work out the tensors $\Pi^{(0)}$ and $\Gamma^{(0)}$. In order to do this, we note that, on the hexagonal grid, the following relations hold:

$$\sum_i e_{i\alpha} e_{i\beta} = 3\delta_{\alpha\beta}; \quad (14.55a)$$

$$\sum_i e_{i\alpha} e_{i\beta} e_{i\gamma} e_{i\eta} = \frac{3}{4}(\delta_{\alpha\beta}\delta_{\gamma\eta} + \delta_{\alpha\gamma}\delta_{\beta\eta} + \delta_{\alpha\eta}\delta_{\beta\gamma}), \quad (14.55b)$$

whereas similar moments containing an odd number of $e_{i\alpha}$ vanish. Together with (14.32), this yields

$$\Pi_{\alpha\beta}^{(0)} = \rho u_{i\alpha} u_{i\beta} + P\delta_{\alpha\beta}, \quad (14.56)$$

where P is found using the ideal gas law relation with the kinetic energy as in Section 14.2. Furthermore

$$\Gamma_{\alpha\beta\gamma}^{(0)} = \rho \frac{c^2}{3} (u_\alpha \delta_{\beta\gamma} + u_\beta \delta_{\alpha\gamma} + u_\gamma \delta_{\alpha\beta}). \quad (14.57)$$

Substituting these equations into (14.54), we obtain

$$\begin{aligned} \partial_t j_\alpha + \partial_\beta \Pi_{\alpha\beta}^{(0)} &= \left(\tau - \frac{1}{2} \right) \Delta t \left[\partial_t \partial_\beta (\rho u_\alpha u_\beta + P\delta_{\alpha\beta}) + \rho \frac{c^2}{3} (\partial_\beta \partial_\beta u_\alpha + 2\partial_\alpha \partial_\beta u_\beta) \right] \\ &\quad + \mathcal{O}(\Delta t^2). \end{aligned} \quad (14.58)$$

In vector notation, this reads

$$\partial_t \mathbf{j} + \nabla \cdot \Pi^{(0)} = \left(\tau - \frac{1}{2} \right) \Delta t \left[\nabla \cdot \left(\frac{\partial P}{\partial t} \right) + \rho \nabla \cdot \left(\frac{\partial \mathbf{u}\mathbf{u}}{\partial t} \right) + \rho \frac{c^2}{3} \nabla^2 \mathbf{u} \right] + \mathcal{O}(\Delta t^2). \quad (14.59)$$

To obtain the last form, we have neglected the term $\nabla(\nabla \cdot \mathbf{u})$, which is small in the incompressible limit as can be seen from the continuity equation. Furthermore, in this limit, the time derivatives on the right hand side are of the order of M^2 – hence we are left with

$$\partial_t \mathbf{j} + \rho \nabla(\mathbf{u}\mathbf{u}) = \nabla p + \rho \nu \nabla^2 \mathbf{u}, \quad (14.60)$$

where we have neglected corrections beyond Δt^2 and $\Delta t M^2$ and where we have used (14.36).

This is the Navier–Stokes equation in the incompressible limit for which $\nabla \cdot \mathbf{u} = 0$. This derivation does not depend on the fact that we have used the hexagonal lattice – it yields the same result for the d2q9 model.

Exercises

- 14.1 Derive from the Navier–Stokes equation that the flow in a two-dimensional pipe forms a parabolic velocity profile. Check that the curvature (i.e. the second

derivative) of the parabola is given by $\nabla P/(\nu\rho)$. Check also that the velocity Δv which is added at each step is related to the pressure by $\rho\Delta v = \nabla P/c$. In the simulation, we usually take the lattice constant Δx and the time step Δt equal to 1. Check that the velocity profile of your simulation matches the result for $\nu = (2\tau - 1)/6$.

- 14.2 [C] Extend the simulation of the parabolic flow by putting an object in the flow. This can be a fixed block. Check your results by inspecting the flow in the channels on both sides of the block to see whether the total flow is the same through the rectangular pipe.
- 14.3 [C] Write a program for droplet formation in a binary system along the lines indicated in [Section 14.4](#).
- 14.4 [C] In this problem we check Laplace's law. In order to do this, you need to identify the droplets and find the pressure drop across their boundaries. To this end you step over all lattice sites and check whether there is a large predominance of one colour. If this is the case, you use either the back-track or the Hoshen–Kopelman algorithm which are both described in [Section 15.5.1](#), in order to identify all sites. Furthermore, you store the largest pressure value in the current droplet.

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